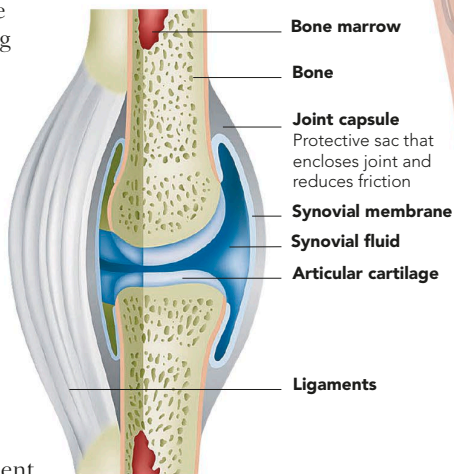


INSIDE A JOINT

The bone ends in a synovial joint are covered by a smooth, slightly compressible tissue called articular cartilage. Surrounding the joint is the joint capsule, which is made of strong connective tissue and is attached to the bone ends. Its delicate inner lining, the synovial membrane, secretes viscous synovial fluid into the synovial cavity to keep the joint well oiled. The fluid also nourishes the cartilage with fats and proteins, and is constantly reabsorbed. Fibrous thickenings of the capsule, called ligaments, are anchored to the bones at each end and prevent unnatural movement of the joint. Muscles around the joint, which are connected to the bones by tendons, provide stability and produce movement.

INSIDE A SYNOVIAL JOINT

A mere film of synovial fluid separates bone ends. There are just 1–2ml ($\frac{3}{100}$ – $\frac{7}{100}$ fl oz) of this liquid in the knee joint.



CARTILAGE AS A SHOCK ABSORBER

The articular cartilage that coats the bone ends in a synovial joint is also known as hyaline cartilage (see p.43). If sudden knocks or vibrations jolt the joint, this cartilage works as a shock absorber to dissipate the force of the impact and prevent jarring damage to the much more rigid bones. In certain joints, the cartilage has especially tough fibres. Examples include the fibrocartilaginous pads, called intervertebral discs, that act as cushions between the vertebrae of the backbone. Fibrocartilage also occurs in the jaw and wrist joints and the menisci in the knee.

SPINAL CARTILAGE

The fibrocartilage discs (blue) between the vertebrae help to stabilize and cushion the spinal column.